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IMAGINING A SOFT AND RELATIONAL SMART HOME

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ABSTRACT

This paper presents three scenario-based speculations accompanied by material explorations of soft IoT for smart homes based on humidity data. *Cycle Lines* tracks and displays weekly patterns of humidity levels shown as colored lines in a woven display, *Relationscape*, a real-time tracker and knitted display of humidity showing the relation between two different households, and *Eco-collective*, embroidery IoT that changes color depending on the humidity levels of different objects in the household. Based on first-person engagements with humidity sensors placed in the authors' homes, they imagine new types of soft IoT devices that sense and display relations to humidity data, suggesting a role for mundane, craft-based IoT for smart homes. They express the relational nature of humidity and how it is tied to well-being in the home, among household members and other human and more-than-human inhabitants, as well as the environmental conditions inside and outside of the home. The soft and craft-based approach to imagining futures of IoT for smart homes has feminist commitments and invites for further problematizing home labor practices and craft activism in the domestic context.

INTRODUCTION

Smart home technologies—such as smart security systems, smart lighting, air conditioning, smart home assistants, and others—are becoming more and more integrated into the homes of Western households. The 'smartness' of these devices is their promised data-driven features and automated functionality (Pink et al., 2017). It is based on the Internet of Things (IoT), a term for electronic devices and sensors that support ubiquitous connectivity to the Internet, connecting to and monitoring things in the real world. The IoT is becoming an important aspect of the household life in relation to things that can be sensed, remotely accessed, and controlled from any computer, mobile phone, or laptop, supporting user needs like security, control, monitoring, and entertainment—predominantly a "command and control" ideology that maximizes information optimization and casts the home as a site for masculine-coded management (Strengers & Kennedy, 2020; Bell & Dourish, 2007).

What is often left out of this research frame is an alternate, feminist path for home IoT that engages with emergent data relations in the smart home as well as the affective and tactile qualities that their materiality could afford. The home is a rich and multifaceted domain, where sentiment, relations, and relationships between people, spaces, and things are constantly being produced and (re)produced over time (Hargreaves et al., 2018; Horst & Sinanan, 2021; Lupton et al., 2021; Pink et al., 2017). Most often, future concepts and products of smart homes are depicted as sleek and efficient, characterized by plastic and hard materials that have quite specific visual and tactile qualities that leave little room for slower (Hallnäs & Redström, 2001; Odom et al., 2014), softer, and more reflective (Sengers et al., 2005) living at home. To imagine alternatives to contemporary IoT, this project draws on previous work exploring how craft materials and practices can invite for attending to embodied, subjective, and affective relations to data that considers personal and

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sociocultural aspects (Devendorf et al., 2020; Moreshead & Salter, 2023; Pérez-Bustos et al., 2019; Rosner et al., 2018). Imagining soft IoT from this perspective reflects an epistemic shift towards producing new domestic technologies rooted in feminist technoscience (Bardzell, 2010; Forlano, 2017; Haraway, 2013), reframing the domestic IoT as an ecosystem taking on new forms to account for multiple (co)inhabitants in the home and their relations in softer ways.

Inspired by these ideas, we set out to explore the following questions: *What other kinds of IoT can we imagine for smart homes, and what kinds of relations do they support or help bring into being? How can we rethink the materiality of the IoT to account to account for other kinds of knowledge and affective experiences?*

To answer those questions, we took as a starting point environmental humidity in homes. Humidity is an easily measurable form of data for the smart home, but it is not always so obvious what different levels of humidity might feel like. It's both arbitrary as a convenient sensor for prototyping, as well as a choice that reflects our situation. Denmark is a country where humidity levels are high and frequently a topic of conversation in relation to the well-being of the home and its inhabitants. High humidity can affect people's health and well-being in their homes by contributing to the development of mold, which can cause serious lung problems.

Using first-person methods (Howell et al., 2021; Ståhl et al., 2021) we lived with humidity sensors in our homes over the course of a month, developing new relations to humidity, and leading to three design spaces for further exploration. We then developed scenario-based speculations for new forms of relational IoT (Blyth, 2005; Carroll, 2003; Tomitsch, 2018). Alongside the scenarios, we used a process of material exploration using fiber crafts, producing propositions for soft IoT to support these new relations.

We contribute with three speculative soft and relational IoT concepts: *Cycle Lines,* tracking and displaying weekly patterns of humidity; *Relationscape*, a real-time display of humidity and temperature between two households; and *Eco-collective,* indicating humidity levels for different objects in a home. We see them as speculative examples of soft and relational IoT, offering opportunities for reflecting on the relationship between domestic technology, labor, and how we want to live in the home.

BACKGROUND

Design and HCI have seen a substantial amount of research into reframing and finding new opportunities for smart home technologies and IoT devices. These include speculative domestic IoT devices for cohousing communities (Jenkins, 2018), unpacking notions of domestic routines (Crabtree & Rodden, 2004), co-speculative investigations of the visions of IoT for the home (Desjardins et al., 2019), exploring the heterogeneity as a framework for domestic design (Aipperspach et al., 2008), studying how the material, technological, and social boundaries between the home and the outside world is constructed, re-made, and curated (Oogjes et al., 2018), as well as how simple interactions and data can be interpreted in and across homes (Gaver & Gaver, 2023; Kurze et al., 2020). In particular, our work is inspired by previous design research concerned with how digital technologies might address environmental issues, specifically design explorations of indoor weather stations and domestic microclimates (Gaver et al., 2013), as well as by research focused on IoT-based home automation systems controlled by weather conditions, with a specific focus on humidity data (Willoughby et al., 2022).

IOT IN THE SMART *HOME*

In their chapter in the book *Making Homes* (Pink et al., 2017), Cox and Buchli define the home as the "*heart of the most seemingly mundane spaces and experiences — the site of activities such as eating, washing, raising children, and loving,*" but also as a space of negotiation, resistance, and oppression—where relationships are made and undone. Vacher has argued "*a home is something you make*" and that people "*make themselves at home*" through an active creation of relationships between people and materials (Vacher, 2010). Desjardins et al. define the home as a complex, nuanced framework for everyday life that has many facets: a constantly changing structure made up of people, things and processes. To these ends, design research has increasingly focused on understanding the complex ways in which interactive technologies are embedded in the home, and their importance to the domestic experience across different forms and constellations of 'home' (Aipperspach et al., 2008; Chatting et al., 2017; Desjardins et al., 2019; Oogjes et al., 2018; Jenkins, 2018).

Critical and speculative approaches to smart home technology has addressed how it can support reflection, address emotion, and promote well-being in the home is central to envisioning better forms of domestic technology (Gaver et al., 2013), Rather than being directly informative to build knowledge of the home, indirect means can "*facilitate users' access to the heterogeneity, ambiguity, and complexity of the world within which the home is situated*" (Michael & Gaver, 2009). This slower, more contextual, and emergent knowledge can offer reflective insight into how we would rather live (Hallnäs & Redström, 2001; Sengers et al., 2005; Odom et al., 2014). These angles on what a smart home can and could become inflects our research,

as we develop different understandings of what living in a smart home might be like.

CRAFT AND SOFT IOT: A FEMINIST TAKE ON SMART HOMES

In reimagining the smart home, we hope to redefine what a smart home feels like. We envision new forms of IoT products for smart homes: soft, colorful, malleable, playful, and ambiguous, which invite for a broader understanding of well-being in the home—a complex interrelation that is distributed among the many actors that co-exist and interact in a domestic context. This includes relations among data, environmental conditions, the residents' well-being, their felt somatic experiences, social worlds: the well-being of the home as a whole. We refer to this *Soft IoT*, an approach that reflects an epistemic shift towards domestic technologies rooted in feminist technoscience (Bardzell, 2010; Forlano, 2017; Haraway, 2013). Feminist HCI emerged in the early 2010s based on constellation of feminist values such as pluralism, participation, advocacy, ecology, embodiment, and self-disclosure (Bardzell, 2010). Many of these values have been explored through design research projects focused on combining craft, yarns, and computation such as (Devendorf et al., 2022; Pérez-Bustos et al., 2019; Rosner, 2018; Rosner et al., 2018; Tsaknaki, 2015; Tsaknaki & Vallgårda, 2023; Nabil et al., 2019; Tan et al., 2016). To explore the design space of soft IoT in smart homes, we began with fiber craft practices like weaving, knitting, and embroidery because their semiotic-material practices place the body into continuity with cultural memory and undervalued feminized forms of craftwork (Moreshead & Salter, 2023; Rosner, 2018).

Textile crafts like knitting, embroidery, weaving or knotting have been used before as ways to (re)materialize and defamiliarize relations to different forms of narratives and data (Devendorf et al., 2022; Muehlbradt et al., 2022; Pérez-Bustos et al., 2019; Tsaknaki et al., 2024). In using fiber craft practices for speculating and materially exploring the space of soft IoT, we drew upon theorizations of fiber crafts as a feminist approach in arts, activism, and design (Moreshead & Salter, 2023). For example, researchers have previously used fiber arts to explore alternative histories of contemporary data storage systems, such as Haring, who explored the long lineage of knitting as a medium for information encryption (Haring, 2011), as well as Rosner et al, who have conducted a series of workshops in which participants "weaved core memory" and through that aimed to bring back to life accounts of forgotten legacies and bodily labor of feminized craft workers to computing (Rosner et al., 2018). Imagining soft IoT for smart homes through the knitting, weaving and embroidery aims to articulate feminist notions of the home as a site of domestic labor by using material practices that have traditionally been both explicitly and implicitly coded as feminine.

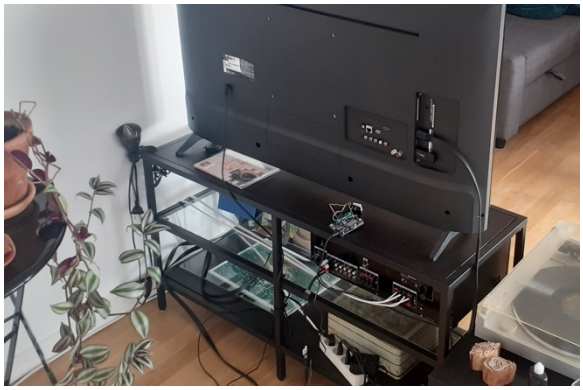




Figure 1: Humidity sensor placements in the "studio" and living room.

FIRST-PERSON ENGAGEMENTS WITH DOMESTIC HUMIDITY

Our research process started with first-person reflections on using humidity sensors in our own domestic contexts. Following previous design methods (Benabdallah et al., 2023; Howell et al., 2021; Ståhl et al., 2021), we recorded our observations and experiences in order to share them among a research group. We placed humidity sensors in our homes to not only engage with humidity, but also to reflect on how the sensors can spark discussions around humidity among members of our households (Figures 1-3).

We used a custom-built Arduino-based sensors to measure humidity in each of our households. It displayed both temperature and humidity levels, updating every 5 seconds, and uploaded this data to a remote server for later reflection. Each author placed the sensor in different locations, based on our initial thoughts and experiences on humidity and well-being at home: The first author placed their sensor on a kitchen counter, where bottles were sterilized and milk prepared for a newborn baby. The second author placed their sensor in a room called "the studio," a spare room where work, play music, and do art and craft hobbies. Finally, the third author placed their sensor in a

bedroom to track how humidity levels changed overnight with the whole family sleeping in the same room. Over one month, each author took photos of sensor values and wrote down reflective notes at least once per day. Sometimes, we changed the sensors' placements, moving their positions in the same room or to another room, based on our experiences and reflections during the study.

During this study we met four times to collectively reflect on the following questions: *How do we relate to humidity in the home, taken broadly, and to humidity data more specifically? Does the sensor data differ from or accurately reflect our own experiences and sensations of humidity?* These questions guided our conversations, which ended up focusing on tracing the discrepancies and differences between our felt experiences and the sensor readings, and how our views and preconceptions of humidity—including how it affected our sense of well-being at home—might have shifted through engaging with the sensor.

This process revealed that what the sensor depicted as a simple numerical value of humidity, was experienced quite differently by different people at different times. Paying close attention to how we felt humidity in our homes, combined with becoming more attentive to the sensor data, created new forms of relating to environmental conditions in the home—and to temperature and humidity in particular—that hadn't existed previously.

We found three areas to further explore from the study:

1. *Sensory representation*—more mundane, more ambiguous, more bodily, and more interpretative displays for humidity data
2. *More-than-human well-being*. How humidity contributes to well-being in domestic contexts for both residents and non-human bodies
3. *Distance and difference.* Understanding the home in relation to weather or to other locations in the world.

Below we present how those three design spaces emerged through our engagements with the sensors, accompanied by notes taken from our reflections.

EXPANDING SENSORY REPRESENTATION

All of us reported, in different ways, that seeing the value on the sensor was limiting. The experience of what it means to see that the humidity in the specific room at this moment is "24,3" for example, often bears no relation to the perceived, felt experience of humidity. The third author noted one day that "*even though the sensor now shows the value of 21 for humidity, which is the same number it showed last night, around that time, the windows are moist and foggy from the inside, and I also feel the air thicker and more humid*". She wished



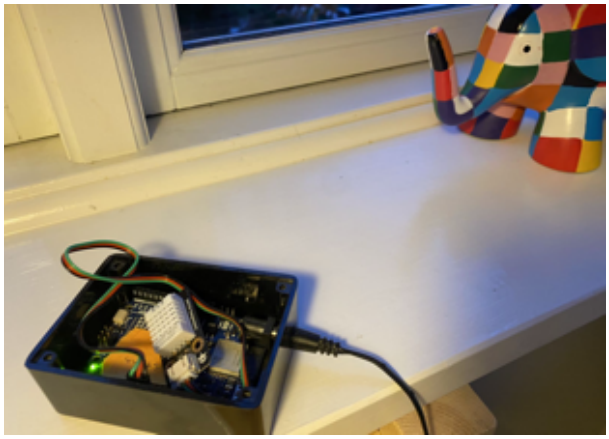


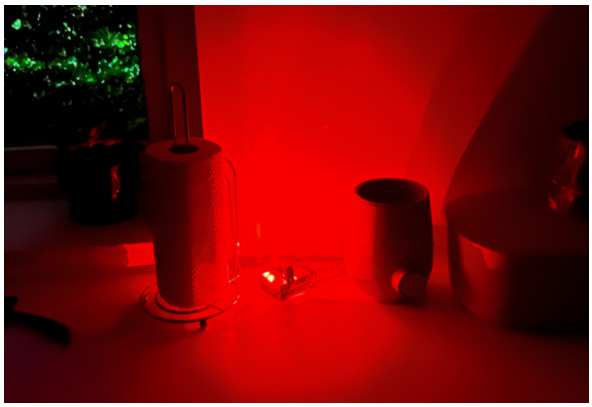
Figure 2: Humidity sensor placements in the bedroom and a living room.

for something that would show more nuance in the humidity values in relation to surrounding spaces, objects, home infrastructure, windows, the temperature of the room, or to how dry or moist the air in her lungs feels when she breathes in.

Through her first-person engagements with the sensor, the same author noted that she wished for another kind of display of humidity in the home for her child, who she would like to engage with the sensing of the home in other ways, rather than relying only on numerical values as the ground truth. These reflections on how to get beyond the valorization of data and numerical displays, resonates with the HCI research on self-tracking as a practice of cultivation rather than control (Jenkins et al., 2021; Boer et al., 2020) , as a phenomenological and bodily grounded practice (Homewood & Vallgårda, 2020), and as a social practice extending beyond numerical systems (Lupton, 2014).

MORE-THAN-HUMAN HUMIDITY ENTANGLEMENTS

Discussing our experiences with humidity sensing, it became apparent that humidity and well-being at home went beyond considering only human habitats. For example, the second author noted in her reflections that becoming attentive to the humidity levels in the studio



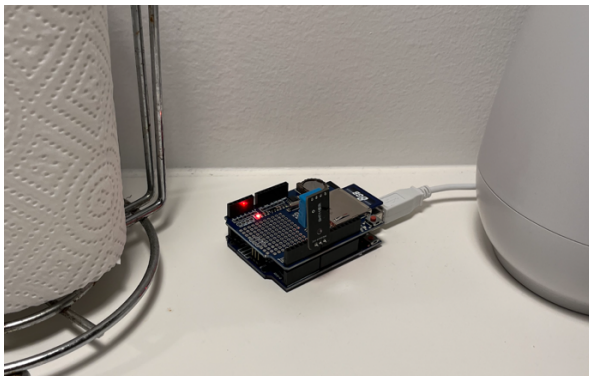


Figure 3: Humidity sensor placed on the kitchen counter.

space sparked discussions among her and her partner on the well-being of their musical instruments (guitars and an electric bass) cohabiting in the same room. They gradually started noting how their experiences of the changing humidity levels could be perceived through dry hands or skin, for example, prompting them to wonder how the instruments responded to change: "*On a very humid day, when we felt an increased moisture in the home and also saw this depicted on the sensor screen, we started worrying about the well-being of the instruments and started searching online for what is a 'healthy' humidity range for the wood of a guitar. We wished to have a humidity sensor close to or attached to the body of the instruments*." Likewise, the other authors also reported becoming more attentive to houseplants and their well-being in relation to different humidity levels at home, compared to how humans experienced humidity levels.

HUMIDITY AS RELATION TO PLACE(S)

The pilot study also revealed that we often experience and relate to humidity in the home by comparing it with the environmental conditions outside. For example, all authors noted how discussions with their co-habitants (variably partner, wife, husband, baby, and daughter) about the numerical humidity values inside most often led to further discussions on weather conditions outside and vice versa: weather conditions outside the home led to an increased interest and more discussions around the humidity levels inside the home. From a small and previously inconsequential fact of the weather, humidity became a matter that mattered to residents' routines, by visiting the weather forecast app on their mobile phone to check the humidity levels in Copenhagen and in other countries or cities where they spend their summer holidays or where their families live.

SPECULATING SOFT IOT FOR HUMIDITY

Humidity data changes slowly, and so we needed to engage with it using a medium that is also slow. To match this, we chose to explore making home goods from thermochromic fabric. A common characteristic of thermochromic materials is that they have a slow response to heat, meaning that it takes some time to change from one state to another—an advantage in making soft, ambiguous, and slow displays of humidity data. Several design researchers have previously explored thermochromic materials—paints and yarns, alone and in combination with textiles—to create alternative forms of displays embedded in furniture or wearables (Redström et al., 2005). Some examples include a tablecloth that invites for reflection through color-change (Gaver et al., 2006), woven and crochet fabric swatches as soft displays for clothing (Devendorf et al., 2016), or speculative home decoration to enable reflections on home water consumption (Nabil & Kirk, 2021).

In our case, we used a cotton thermochromic fabric that changes color from light blue to white at 30°C. We used it both as a material directly as well as cut into strips and turned into spools of thermochromic yarn. We crafted different probes with combinations of thermochromic and regular yarn and tested the color-change output through heating pads and conductive thread that warmed when current was applied. These led to three material explorations representing different ways of materializing aspects of the design spaces from the first-person engagement with the humidity sensors.

Alongside this material exploration, we developed three scenarios aiming to both capture our experiences of using the humidity sensors, as well as support us in further imagining soft IoT sensing and actuating devices for humidity. The scenarios became a means to explore the design spaces, helping us to contextualize and understand what we were making. Developing scenarios aims to document and communicate design ideas and evoke reflection on the content of the design work (Blyth, 2005; Carroll, 2003; Jenkins et al., 2020).

These scenarios imagine new forms of humidity IoT devices for smart homes, and support conversations relating humidity to the well-being of the household. Together, the scenarios and their partial materializations aim to offer different ways of attending to the affective, embodied, and subjective relations to humidity data. In each concept developed, mapped to a different scenario,

the thermochromic yarn is used to imagine how humidity data aggregated through a humidity sensor can be mapped as output to an ambiguous soft display. *Cycle Lines* is a weekly tracker of humidity levels, depicted as colored lines in a woven display inside the home, *Relationscape* is a real-time knitted display of humidity, showing the relation between humidity levels of two different households, and finally the *Eco-Collective* is an embroidered display that changes color depending on the humidity levels of different objects in the household. Below, we present each concept through a brief overview, followed by the scenario and then by a description of the material exploration.

CYCLE LINES

Cycle Lines is a way to bring conversations around humidity into the household, without focusing on numerical representation of data. Instead, it draws attention to the feelings, emotions, and well-being that different humidity levels inside and outside the home. Cycle Lines updates the current day's fiber row every ten minutes, using the average humidity for that day as the final color of the row before moving to the next row at midnight. The Soft IoT device displays humidity data information from the last seven days, showing lines of different colors representing shifts in humidity over time. The scenario follows:

Christian gets home with his 5-year-old daughter, Sally. They are both soaking wet from the rain pouring down outside. Sally is eager to get her snowsuit off. Christian helps her with the jacket. Afterwards he makes a coffee for himself and a plate with some sliced fruit, banana and apple and pours a glass of milk for Sally. When they both sit at the table, it feels a little calmer. Now they can start to ground themselves at the home environment and look around the house. The windows are a bit moist and foggy due to the rain still pouring outside. Luckily it feels dry and warm inside. They look at the Cycle Lines standing in a frame on the kitchen counter. A woven line is slowly changing its color from blue to white. Sally contemplates "it looks really nice with the changing color, don't you think dad?" "Yes," Christian replied, "*next to the grey and the green, it creates a nice row of colors.*" Sally looks at the humidity forecast frame: "*Look at the frame, dad! Yesterday we also had lines of white, grey and green, just in another order. Green first, then grey and then white*." *Christian looks at the frame again, and replies:* "*That's a good observation. It creates a nice pattern, I think. We might get a whole week of it, but let's see*."

The humidity IoT frame displays the humidity changes in the home as well as the humidity outside during a week. These are represented by different woven lines at different places in the frame creating an everchanging color pattern to create felt knowledge of the humidity effect in the home.



Figure 4: The *Cycle Lines* soft IoT is made of hand-woven combinations of thermochromic and regular yarns. Testing the color change through a heat pad attached to the back of the frame.

We developed this prototype (Figure 4) through hand-weaving combinations of thermochromic and regular yarns on a weaving frame to create a soft, inviting material. Thermochromic areas are alternating with non-thermochromic ones, creating lines and sections that can change their color when heat is applied to them, and others that will not be affected. We conducted tests focused on different temporalities of heating the thermochromic areas through heating pads attached to the back of the frame.

We imagine it to be placed on a table surface, or on the wall as a decorative piece. Data from humidity data aggregated from two different resources (e.g. different rooms in the house, inside/outside of home) could be mapped to the different thermochromic areas, creating intricate patterns of visualizing humidity data. The patterns emerging can change slowly over time and invite for deeper engagements with the data, to understand it in new ways.

THE RELATIONSCAPE

The *Relationscape* is soft IoT invites for relating to humidity data through connecting to another person and/or another location. When setting it up, the people involved must visit the *Relationscape* website and register the location of their sensor. The *Relationscape* aggregates and projects the humidity data of one home with a home somewhere else. These are depicted in different areas on its knitted thermochromic display. Its intricate color patterns slowly change throughout the day as humidity data is updated and applied to the knitted structure in real time. We imagine it to be hanging from the ceiling as a mobile or being placed on the wall, as a home decoration. It invites for a slow and ambiguous engagement with experiences of humidity in the home, prompting reflections on how weather conditions and well-being are relational and can connect to other places and the life of significant others living further away. The scenario follows:

Jenny got home a few minutes after 6pm. She removes her coat and shoes and gets a glass of water before sitting on the couch. She realizes how wet her hair and clothes are from the rain, leaving wet marks on the fabric of the sofa. The rain is pouring down outside. It had been like this for the last week at least, and the weather forecast showed no signs of sunshine. She looked at the Relationscape, a knitted mobile structure hanging from the ceiling next to the sofa. It had an intriguing color pattern—an area with blue tones and another area with pink tones. She wrapped herself in a blanket. "I wish I was with Alice in Australia where the warm, dry weather is" she thought to herself. She had visited her sister Alice in Australia three months ago. It was summer in Australia and winter in Europe where Jenny lived. They were very close, even though Alice had been living on the other side of the earth for two

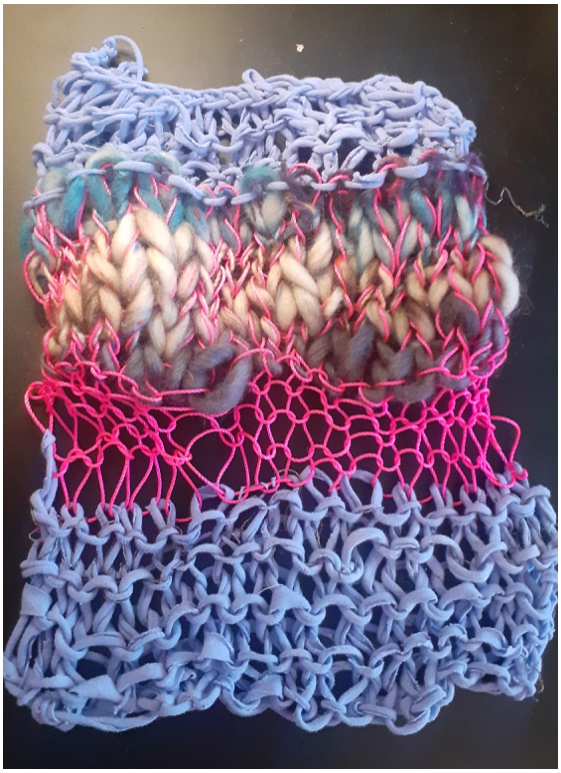
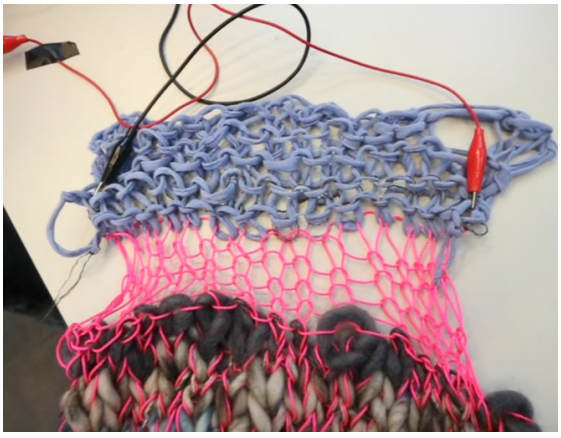


Figure 5: *Relationscape* prototype made of-hand knit combination of thermochromic and standard yarns.



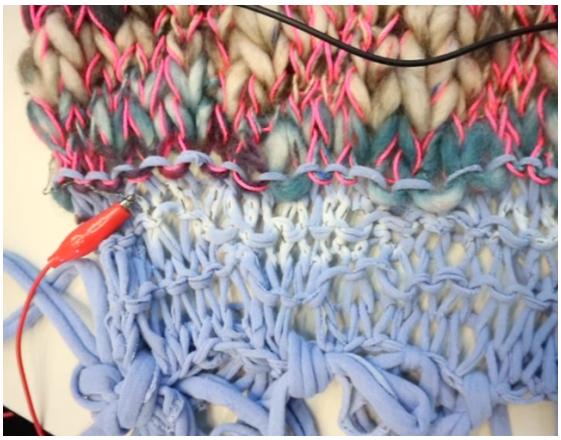


Figure 6: Testing the color change through applying current to the conductive thread stitched and knitted into the structure.

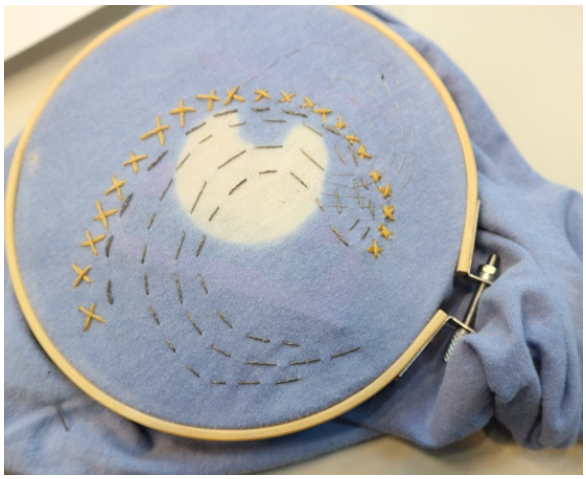
years now. Last time she visited, they decided to set up a humidity sensor in Alice's home, so that Jenny could connect to her in real-time in a way that wasn't very intrusive. The different time zones made it difficult to connect on Zoom or over the phone: when Jenny was awake and at home, Alice was asleep and the other way around. The Relationscape gave the sisters a way to stay in touch in a more subtle way, giving Jenny a more direct connection and feeling to the home of Alice. "I will write Alice and send her a picture of my rainy day" she thought to herself. She takes a picture of the gray skies outside and matches it with one of the current color of the Relationscape, sending them to Australia. She feels a bit better with Alice in mind and with the blanket starting to warm her body.

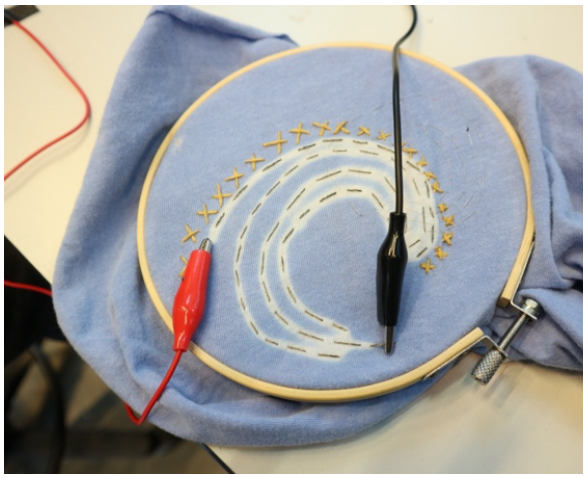
The *Relationscape* (Figure 5) uses knitting as a craft technique. It was explicitly made to have a loose-knit structure and be airy, spider-web appearance and loose thread aesthetics. The thermochromic yarn was used in combination with non-thermochromic yarns of different quality, thickness and color. In one of the thermochromic areas, conductive thread is woven together with the thermochromic yarn, and in another area, conductive thread is stitched along the surface of the thermochromic yarn. We tested those two different ways of heating up the thermochromic cotton yarn, when applying current to the conductive thread knitted along with it, aiming to find the most efficient way of avoiding short-circuit (Figure 6). We found that when the conductive thread is stitched on the yarn's surface, results are better in terms of a clear mapping of input and output. This solution is also more robust and less prone to lead to short-circuit.

THE ECO-COLLECTIVE

The *Eco-collective* is a humidity IoT device that connects human and more-than-human inhabitants in the household. Through an embroidered soft display, it visualizes well-being in terms of humidity as relations between human nonhumans in the domestic context. The embroidered patterns represent a holistic approach of well-being in the home, by considering it an entangled relation of the well-being of inhabitants, plants, guitars or a piano. They change color depending on whether humidity enter or exit predefined humidity thresholds or ranges.The humidity data is expected to be updated every morning at 3am. The scenario follows:

Sebastian and Jacob are making breakfast on a Sunday morning after coming back from visiting their friends in their summerhouse. Sebastian looks at the Ecocollective, an embroidery hanging on the wall in the kitchen. It has a stitched lines and patterns depicting four different plants, the two guitars and the piano: the guitars are blue, one plant is dark grey, and another one is orange. "Look, some of the stitches representing the plants have turned dark grey, Jacob." Jacob looks





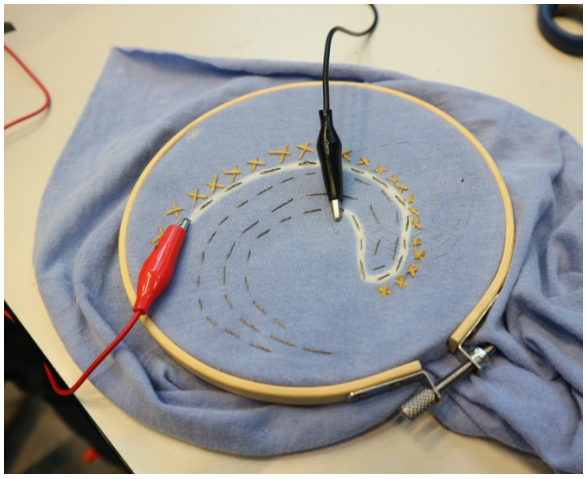


Figure 7: The *Eco-collective* embroidered soft humidity IoT: testing the color change from blue to white, around the stitches with conductive thread, when heat is applied.

at the embroidery and responds: "Ahh yes, I think it is the fig tree. It did not do that well before we left, but now it seems that is doing really poorly. Should we move it to a sunnier spot?" Sebastian takes a can of water and goes down the hallway to the living room, where a large fig tree is standing in a corner of the room. He moves the fig tree closer to the window and

waters the plant. As he walks back down the hallway, he makes a stop at the studio and taps on the wooden body of one the guitars, placing his ear closer to the wooden surface. Then he walks back down the hallway. The wood of the guitar sounds fine to him.

The material probe developed for tactile imagining and experiencing the *Eco-collective* (Figure 7) is an embroidered thermochromic display. Here, thermochromic fabric was used for stitching conductive, regular and thermochromic thread, exploring different temporalities and patterns of color change, when applying different levels of heat to the materials. Testing this prototype with current applied to the conductive thread for heating it up, showed intriguing paths forward. We imagine the *Eco-collective* to be a wall decoration or turned into a decorative pillow. The embroidery sample depicted in Figure 7, changes color and turns from blue to white, around the conductive thread stitches.

DISCUSSION

These three smart home concepts reflect a commitment towards developing new forms of embedded and embodied knowledge in the home. As with the post-phenomenological turn in self-tracking technologies (Homewood et al., 2019; Boer et al., 2020; Jenkins et al., 2021; Lupton, 2014), where people track themselves beyond numbers, to know themselves in new and more reflective ways, we see these technologies as extending notions of the body into affective space and relation. In this view, the home becomes part of the sensory apparatus of residents, rather than a sensing entity that represents itself in the abstract for consumption by residents.

In this vein, we see the three developed scenarios accompanied by the material probes as inviting for imagining new types of soft IoT devices that sense and display relations to humidity data, suggesting a role for soft and craft-based IoT in smart homes. We argue that the visual and tactile aesthetics of the material explorations tie the relational nature of humidity to concepts of well-being in the home. Together, they illustrate how a prosaic and unassuming topic like humidity can bind together household members, other humans, more-than-human inhabitants, as well as the environmental conditions inside and outside of the home in new ways—together underscoring the home as semi-permeable and porous space.

RELATING TO (ENVIRONMENTAL) DATA

The first way the scenarios and material probes that we developed operate is to help rethink how we experience *data* in smart homes. Together, they suggest thinking more broadly about how we experience environmental data through IoT devices in domestic contexts. They explore emergent, relational, and open-ended engagements with data to move away from displaying humidity values as numerical, towards depicting humidity as a material participant in the home—something that itself relates to bodies and materials in different ways. These examples question sensing devices that offer an external "ground truth" and highlight the experiential, affective and interpretational approach towards sensor readings and sensor interpretations (Desjardins et al., 2020; Alfaras et al., 2020).

We also see this work as postulating *another form* of smart home. Smart homes have long been the domain of predominantly men (Chetty et al., 2007; Bell & Dourish, 2007), where "smartness" is used to pursue goals like command, control, security, and monitoring. The soft IoT object in this project reflects a different orientation to what homes might be if they become smart. Often, ubiquitous computing implicitly assumes that information should be presented to us in clear ways that arrive at the right time in the right context to assist us in living (Crabtree et al., 2012). Similarly, smart home information screens often take the form of dashboards that give a "birds-eye" view of the home (e.g. technology platforms like Home Assistant[1]) that contain all the information one could need. The data representations in these soft IoT concepts, however, imagine that residents need to perform active sensemaking of what is happening—there is not an easy avenue towards passive awareness (Rogers, 2006). When not being attended to, the objects fade into the background, becoming décor.

ARTICULATING DOMESTIC COMPLEXITY

The second way these probes operate is to provide a perspective for articulating *relations* in smart homes. These scenarios attempt to surface the complex reality of everyday life, where environmental data sensing is not neutral and objective, and the home is not discrete from its environment, unrelated to what goes on around it. The home, its inhabitants, and the experience of the home are constantly changing because of the interactions with the environment and environmental data. Naturally, different types of humidity sensors and different sensor placements will affect the sensor readings, as others have explored when engaging with

[1] https://www.home-assistant.io

sensors diffractively (Sanches et al., 2022). Taking a cue from this work, our concepts envision modes of relating to and (re)making experiences in the home by using humidity data in intimate and affective ways. These concepts reflect how we might like to live: they give form and articulate relations between household inhabitants, and the things making up the home and the environment. Developing these kinds of relations in 'smart homes' lets us imagine other forms of homes — sensing homes that are sensitive and concerned with the well-being of human as well as more-than-human needs.

MAKING HOMES BEYOND 'SMART'

Finally, these prototypes invite us to imagine *making homes* in ways that both extend and upend traditional homemaking. Using fiber craft as materials for IoT systems, we offer a different idea of what smart home technology might look and feel like, as well as what domains smart home technologies might operate in. Thinking through hands, yarns and soft materials in this research space proved generative for escaping more rigid notions of security, control and efficiency that are often related to futures of IoT and instead focused on values of ambiguity (Gaver et al., 2003), playfulness, softness, and slowness (Odom et al., 2014; Hallnäs & Redström, 2002). The crafted soft IoT approach to humidity data brings forward the importance of home aesthetics as well as the home as a personal space that creates constantly changing relations.

The craft practices that produced these material speculations themselves reflect a commitment to home *making*. In this frame, imagining the smart home as something that is made to purpose and fit to individual wants and needs helps to support the idea of domestic worlds that are customized to support and sustain relations that are local to the home. Craft materials as a basis for the home IoT can offer a starting point for other bottom-up smart home practices, where people might build their own responsive systems for their own individual needs (Claus et al., 2023; Moreshead & Salter, 2023). Different people in different homes will of course have correspondingly different interests in what it is that a smart home can or should do for them, and it makes sense that off-the-shelf solutions from large technology companies might not fit everyone. If we think of the home as an extension of the self, as above, and imagine new forms of smart home cultures, then personalization, expressiveness, and the right thing for the right place is essential.

A SOFT IOT AGENDA

Soft and craft approaches to IoT products for smart homes could also invite for reflect and reframe the fact that homes have historically been sites of domestic labor, especially for women. Working at home and taking care of the home, kids, the household, has been both an undervalued and an unpaid form of labor (Federici, 2021). One might wonder whether there is an irony in developing stereotypically femininized forms of craft objects for technical devices to (re)claim ownership of the domestic sphere. However, because homes have historically been spaces where practices like knitting, weaving and embroidery were taking place, it makes sense to use craft practices to explore the space of designing soft IoT. We imagine that this soft IoT can spark modes of conversations and awareness connected to feminist approaches to smart homes and IoT devices, while at the same time problematizing domestic labor practices. We plan to explore these threads in future work.

CONCLUSION

This paper presents three scenario-based speculations accompanied by material explorations of soft IoT for smart homes based on humidity data. *Cycle Lines* tracks and displays weekly patterns of humidity levels shown as colored lines in a woven display, *Relationscape*, a real-time tracker and knitted display of humidity showing the relation between two different households, and *Eco-collective*, embroidery IoT that changes color depending on the humidity levels of different objects in the household. Based on first-person engagements with humidity sensors placed in the authors' homes, they imagine new types of soft IoT devices that sense and display relations to humidity data, suggesting a role for mundane, soft and craft-based IoT for smart homes.

They express the relational nature of humidity and how it is tied to well-being in the home, among household members and other human and more-than-human inhabitants, as well as the environmental conditions inside and outside of the home. The soft and craft-based approach to imagining futures of IoT for smart homes has feminist commitments and invites for further problematizing domestic labor practices and craft activism in the domestic context. Future development of IoT for smart homes should recognize and account for that relationality, considering homes as something that are in constant making and re-making.

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